ECA 1011 – MICROPROCESSOR AND MICROCONTROLLERS

Name : S.Harshavardhan

Reg No : 192524106

EXPERIMENT – 3:-

Multi-Byte Data and Array Operations Using 8086 Indirect Addressing

1. Objectives

1. To understand the concept of indirect addressing in the 8086 microprocessor.
2. To perform multi-byte data operations using memory locations.
3. To access array elements using index and pointer registers.
4. To perform data transfer and arithmetic operations on array elements.
5. To understand the use of registers such as SI, DI and BX for memory addressing.

2. Components Required

1. 8086 Microprocessor.
2. 8086 Microprocessor Trainer Kit.
3. Personal Computer (PC).
4. Assembler Software (EMU8086).
5. Power Supply.
6. Connecting Cables.

3. Procedure

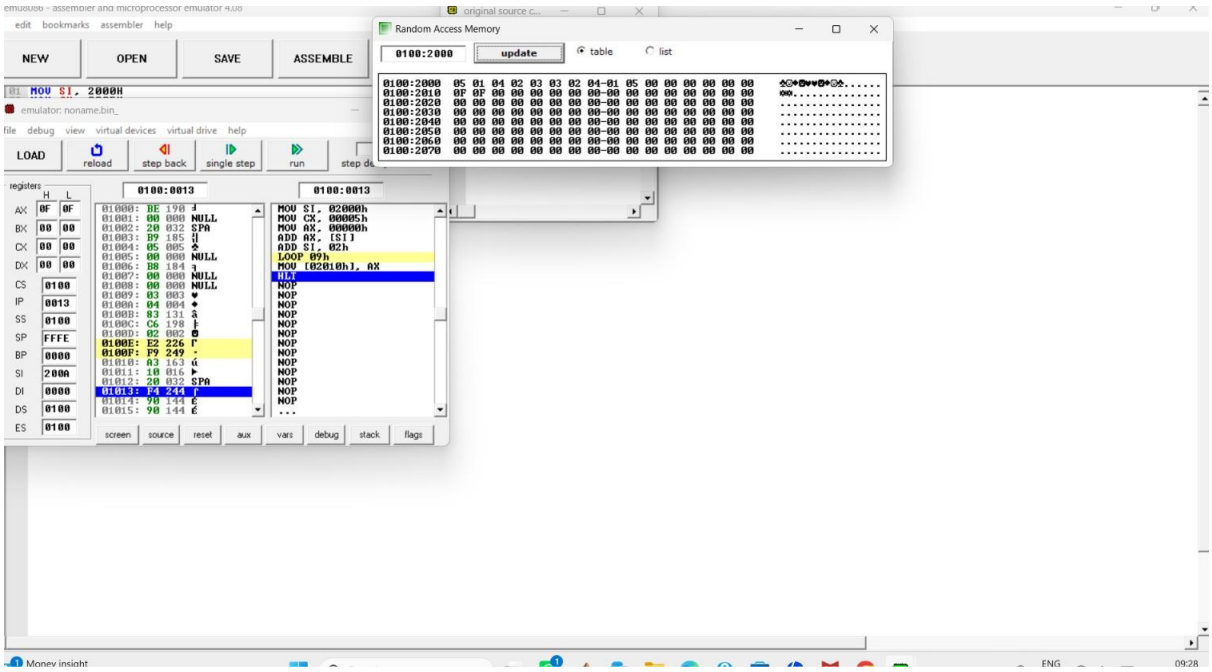
1. Switch ON the 8086 microprocessor trainer kit.

2. Open the EMU8086 assembler software or the available 8086 programming environment.
3. Write the assembly language program for multi-byte data and array operations.
4. Initialize the data segment register (DS).
5. Store the required data and array elements in memory locations.
6. Use indirect addressing registers such as SI, DI or BX to access the memory locations.
7. Perform the required operations on the array elements.
8. Execute the program step by step.
9. Observe the contents of registers and memory locations.
10. Verify the output and record the results.

1. To find the sum of N 16-bit numbers stored in a word array.

8086 program

```
MOV SI, 2000H
MOV CX, 0005H
MOV AX, 0000H
NEXT: ADD AX, [SI]
ADD SI, 02H
LOOP NEXT
MOV [2010H], AX
HLT
```



2. To add two 64-bit numbers byte by byte with carry propagation

2.8086 Program – Addition of Two 64-Bit Numbers

MOV SI, 2000H

MOV DI, 2008H

MOV BX, 2010H

MOV CX, 0008H

CLC

NEXT: MOV AL, [SI]

ADC AL, [DI]

MOV [BX], AL

INC SI

INC DI

INC BX

LOOP NEXT

MOV AL, 00H

ADC AL, 00H

MOV [2018H], AL

HLT

